



Semester 6, Spring, 2025-2026
Course CCNA
Lab Static VS Dynamic Routing
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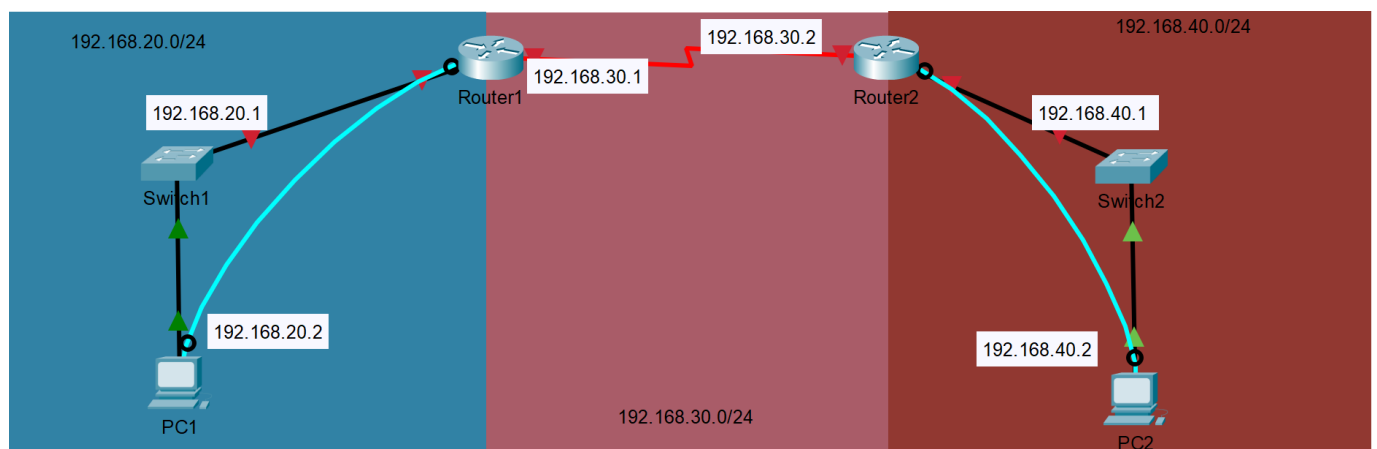
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Objective

The objective of this lab is to connect two separate Lan networks using two routers through a serial connection, configure static routing and dynamic routing (RIP), and verify end-to-end connectivity between hosts.

Network Topology Description



LAN 1

- PC1: 192.168.20.2 /24
- Switch1
- Router1:
 - G0/1 → 192.168.20.1 /24
 - Serial S1 → 192.168.30.1 /24

LAN 2

- PC2: 192.168.40.2 /24
- Switch2
- Router2:
 - G0/1 → 192.168.40.1 /24
 - Serial S1 → 192.168.30.2 /24

Serial network between routers:

- 192.168.30.0 /24

Tools and Equipment Used

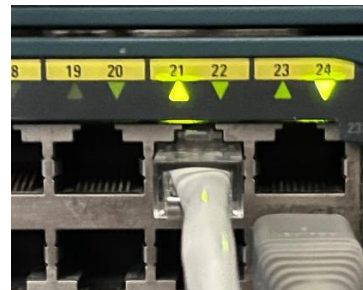
- Two Cisco routers
- Two switches
- Two PCs (Windows OS)
- Ethernet cables
- Serial DCE/DTE cable
- Console cable
- PuTTY terminal emulator

Physical Connections

- I. Connected PC1 to Switch1 and PC2 to Switch2 using Ethernet cable.



- II. Connected Switch1 to Router1 G0/1 and Switch2 to Router2 using Ethernet cable.



- III. Connected Router1 to Router2 using a Serial cable.



Routers Configuration

A console cable was used to configure each router.

- I. Connected console cable from PC to router console port.



- II. Opened PuTTY on Windows.
- III. Selected Serial connection.
- IV. Clicked Open to access the CLI.

Router1 Configuration

Configure G0/1:

```
Router1#configure terminal
Router1 (config)#interface g0/1
Router1 (config-if)#ip address 192.168.20.1 255.255.255.0
Router1 (config-if)#no shutdown
```

Configure Serial interface:

```
Router1#show controllers s0/0/1
Interface Serial0/0/1
Hardware is SSC
DCE V.35, clock rate 64000
Router1#configure terminal
Router1 (config)#interface s1
Router1 (config-if)#ip address 192.168.30.1 255.255.255.0
Router1 (config-if)#clock rate 64000
Router1 (config-if)#no shutdown
```

Noticed that the DCE is on the Router1 serial port, and we assign the clock rate to 64000.

Router2 Configuration

Configure G0/1:

```
Router2#configure terminal
Router2 (config)#interface g0/1
Router2 (config-if)#ip address 192.168.40.1 255.255.255.0
Router2 (config-if)#no shutdown
```

Configure Serial:

```
Router2 (config)#interface s1
Router2 (config-if)#ip address 192.168.30.2 255.255.255.0
Router2 (config-if)#no shutdown
```

PC IP Configuration (Windows)

- I. Opened Control Panel.
- II. Selected Network and Internet.
- III. Opened Network and Sharing Center.
- IV. Clicked Change adapter settings.
- V. Right-clicked Ethernet → Properties.
- VI. Selected Internet Protocol Version 4 (TCP/IPv4).

PC1

IP Address: 192.168.20.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

PC2

IP Address: 192.168.40.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

Static Routing Configuration

On Router1:

```
Router1(config)#ip route 192.168.40.0 255.255.255.0 192.168.30.2
```

On Router2:

```
Router2(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.1
```

show ip route (on Router2):

```
S      192.168.20.0/24 [1/0] via 192.168.30.1
C      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.30.0/24 is directly connected, Serial0/0/1
L      192.168.30.2/32 is directly connected, Serial0/0/1
C      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.40.0/24 is directly connected, GigabitEthernet0/1
L      192.168.40.1/32 is directly connected, GigabitEthernet0/1
yourname#show ip int brief
Interface
```

Notice that the S field (static) is connected to 192.168.20.0, which is the network of PC1.

Testing Static Routing

From PC1:

```
C:\Users\Admin>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:
Reply from 192.168.40.2: bytes=32 time=19ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 19ms, Average = 18ms
```

From PC2:

```
C:\Users\Admin>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time=18ms TTL=126
Reply from 192.168.20.2: bytes=32 time=18ms TTL=126
Reply from 192.168.20.2: bytes=32 time=18ms TTL=126
Reply from 192.168.20.2: bytes=32 time=18ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 18ms, Average = 18ms
```

Result: Successful replies received.

RIP Configuration (Dynamic Routing)

Remove static routes:

```
Router1(config)#no ip route 192.168.40.0 255.255.255.0 192.168.30.2
```

```
Router2(config)#no ip route 192.168.20.0 255.255.255.0 192.168.30.1
```

On Router1:

```
Router1(config)#router rip
Router1(config-router)#version 2
Router1(config-router)#network 192.168.20.0
Router1(config-router)#network 192.168.30.0
```

On Router2:

```
Router2(config)#router rip
Router2(config-router)#version 2
Router2(config-router)#network 192.168.40.0
Router2(config-router)#network 192.168.30.0
```

Verification using show ip route:

```
*Feb 17 13:05:31.879: %SYS-5-CONFIG_I: Configured from console by cisco
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - B
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS le
ia - IS-IS inter area, * - candidate default, U - per-user stati
o - ODR, P - periodic downloaded static route, H - NHRP, l - LIS
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

R    192.168.20.0/24 [120/1] via 192.168.30.1, 00:00:21, Serial0/0/1
C    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.30.0/24 is directly connected, Serial0/0/1
L    192.168.30.2/32 is directly connected, Serial0/0/1
C    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/1
L    192.168.40.1/32 is directly connected, GigabitEthernet0/1
```

Notice that R (rip) is connected to the PC1 network (same on Router1).

Testing RIP

```
C:\Users\Admin>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:
Reply from 192.168.40.2: bytes=32 time=19ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126
Reply from 192.168.40.2: bytes=32 time=18ms TTL=126

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 19ms, Average = 18ms
```

Result: Successful communication confirmed.

Firewall Testing

Enabling Windows Firewall

On one of the PCs (for example PC2), the Windows Firewall was enabled.

Procedure:

1. Opened Control Panel.
2. Selected System and Security.
3. Opened Windows Defender Firewall.
4. Enabled the firewall for the active network profile.

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Private network settings

-  ☒ Turn on Windows Defender Firewall
- ☐ Block all incoming connections, including those in the list of allowed apps
- ☒ Notify me when Windows Defender Firewall blocks a new app
-  ☐ Turn off Windows Defender Firewall (not recommended)

Public network settings

-  ☒ Turn on Windows Defender Firewall
- ☐ Block all incoming connections, including those in the list of allowed apps
- ☒ Notify me when Windows Defender Firewall blocks a new app
-  ☐ Turn off Windows Defender Firewall (not recommended)

Connectivity Test After Enabling Firewall

Try to ping from PC1:

```
Pinging 192.168.40.2 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
  
Ping statistics for 192.168.40.2:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The ping request failed.

No reply packets were received.

Conclusion

In this lab, two independent LAN networks were successfully connected using two routers through a serial cable. Static routing was configured manually and verified through tests. RIP version 2 was then implemented to enable dynamic route between routers.

The lab was performed using real routers, switches, serial cables, and console access through PuTTY, providing practical hands-on experience in hardware-level network configuration. End-to-end communication between hosts in different networks was successfully achieved using both static and dynamic routing methods.